
Thyroid Gland

- Location: Front of neck, below larynx; two lobes joined by isthmus.
- Hormones: Thyroxine, Calcitonin (calcitonin not in syllabus).
- Function of Thyroxine:
 - Regulates basal metabolism.
 - Controls oxidation, growth, development, body temperature, and metabolism of fats, proteins, carbohydrates.
- Iodine Importance: Iodised salt; iodine needed for thyroxine production.
- Thyroid Deficiency (Hypothyroidism):
 - Simple Goitre – Enlargement of thyroid; swelling in neck.
 - Cretinism – Dwarfism, mental retardation (in children).
 - Myxoedema – Swelling of face and hands, sluggishness (in adults).
- Thyroid Excess (Hyperthyroidism):
 - Exophthalmic Goitre – Protruding eyes, rapid heartbeat, restlessness, shortness of breath, high metabolism.

Pituitary Gland

- Location: Below hypothalamus; called "master gland".
- Structure: Two lobes – anterior and posterior pituitary; intermediate lobe vestigial in humans.
- Anterior Pituitary Hormones:
 1. Growth Hormone (GH) –
 - Deficiency in childhood → dwarfism.
 - Excess in childhood → gigantism.
 - Excess in adulthood → acromegaly (overgrowth of face, jaws, hands, feet).
 2. Thyroid Stimulating Hormone (TSH) – Activates thyroid.
 3. Gonadotropic Hormones – Regulate testes and ovaries.
 4. Adrenocorticotrophic Hormone (ACTH) – Activates adrenal cortex.

Posterior Pituitary Hormones

1. Antidiuretic Hormone (ADH) – Controls water reabsorption by kidneys; deficiency → diabetes insipidus (excessive dilute urine).
2. Oxytocin – Stimulates uterine contractions during childbirth, aids milk ejection.

Control of Hormonal Secretions

- Negative Feedback: Hormone secretion adjusted to maintain normal level (e.g., TSH level).
- Positive Feedback: Rare; e.g., uterine contractions during childbirth continue until delivery.